U.S. ARMY

PROTOTYPE RCCTO RAPID DELIVER

HUMAN MACHINE INTEGRATED FORMATIONS PROJECT OFFICE

RCCTO

ARMY RAPID CAPABILITIES AND CRITICAL TECHNOLOGIES OFFICE

Human Machine Integrated Formations (HMIF) Overview

6-7 May 2025

Version 8.1

HMIF SoS Architecture Review

Review Purpose:

To share with Industry the HMIF System of Systems (SoS) Architecture framework, modularization philosophy, boundaries, associated standards, and interfaces of modules to inform Army modernization priorities related to robotics and autonomous systems (RAS).

Industry Collaboration Purpose:

- To assess the detailed description of HMIF software frameworks, and to conduct a collaboration event with Industry as market research for the immediate request to develop control, command, and autonomy software for HMIF Increment 1.
- To obtain industry feedback during the event.
- To enable Industry to self-form and help the Army progress with Human Machine Integration (HMI).

Disclaimer: Vendors participating in this event are not in any advantaged position for any competitions or awards.

HMIF Overview

Mission Statement: To accelerate the fielding of a robotic formation that leverages machines to offload risk from Soldiers and provide additional information for decision-making in Armored and Infantry Formations.

Description: Provide HMIF sets (Armored and Infantry) for rapid prototyping and leave-behind capability in U.S. Army units for learning, iterative refinement, and operational employment.

Characteristics:

1. Infantry HMIF Platoon (PLT) to an Infantry Brigade Combat Team (IBCT).
2. Armored HMIF Platoon to an Armored Brigade Combat Team (ABCT).
3. Each formation will include ground and air systems and enablers to aid human decision-making to find, fix, and engage enemy targets.
4. Prototype development supports existing and future robotic and autonomous Programs of Record (PoR) by mitigating risk associated with the common architecture, communications, network, and mitigating safety risks hindering operational employment.
5. In addition to ground platforms, HMIF will be integrated with Unmanned Aircraft System (UAS), enablers, and a variety of payloads from existing PoRs or developed and transitioned to PoRs.

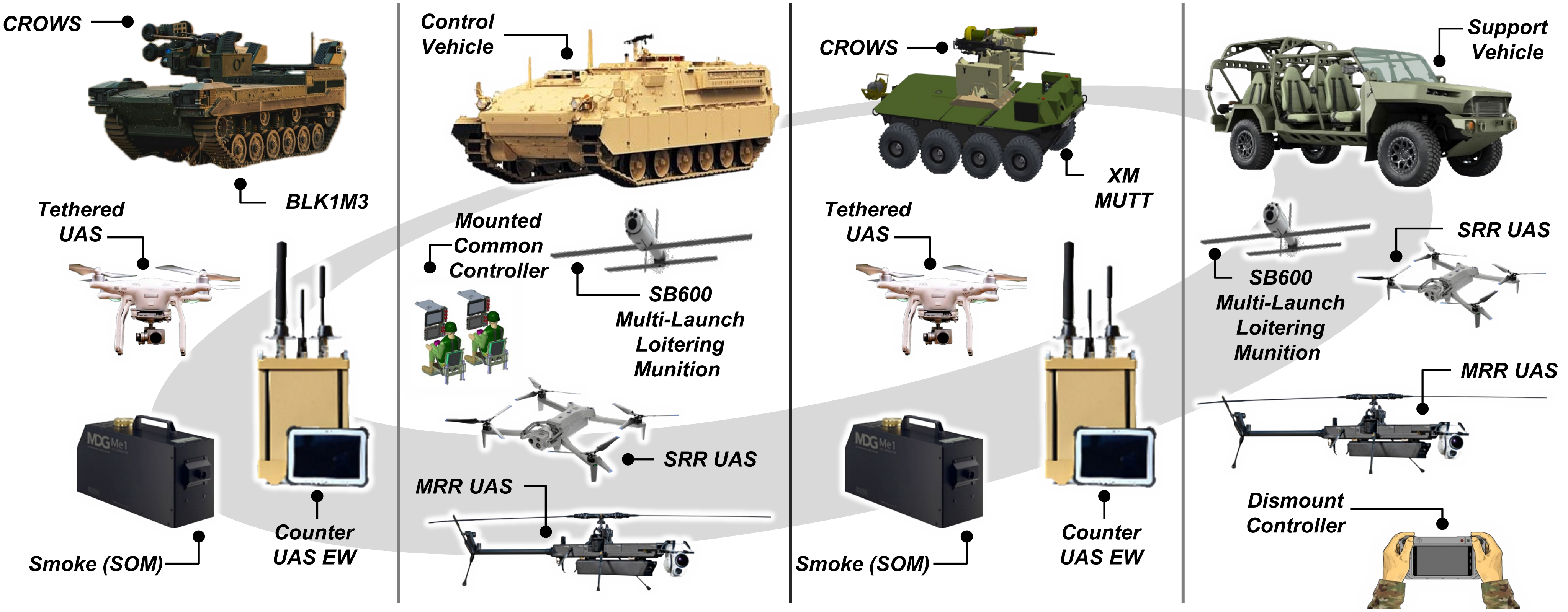
"Think big, start small, and go fast!"

HMIF Increment 1

HMIF-Armored (HMIF-A)

HMIF-Infantry (HMIF-I)

Baseline Configuration	***Soldier Operated Control Vehicle (CV)***	***Baseline Configuration***	***Soldier Operated Support Vehicle (SV)***



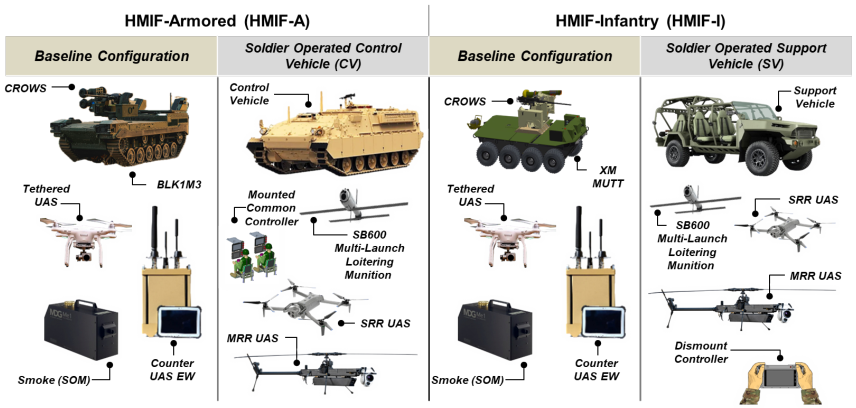
- **Aligns Science and Technology from three AFC DEVCOM centers (GVSC, AC, C5ISR) to integrate and harden prototypes.**
- **Expands defined architecture and common compute across platforms enabling Modular Open System Approach (MOSA).**
- **Enhances communications of HMIF platforms, payloads, and Command-and-Control (C2) nodes.**
- **Further develops teleoperation of unmanned ground vehicles to enable autonomous behaviors.**
- **Upgrades to platforms and payloads are tested for safety release to Soldiers.**
- **Development of common software-defined mounted and dismounted controllers to reduce the number of devices and complexity.**

AC - Armaments Center
AFC - U.S. Army Futures Command
C5ISR - Command, Control, Communication, Computers, Cyber, Intelligence, Surveillance and Reconnaissance Center
CROWS - Common Remotely Operated Weapon Station
DEVCOM - U.S. Army Combat Capabilities Development Command
DUST - Deployable Utility Support Trailer
EW - Electronic Warfare
GVSC - Ground Vehicle Systems Center
HMIF - Human Machine Integrated Formations
MRR - Mid-Range Reconnaissance
SOM - Screening Obscuration Module
SRR - Short Range Reconnaissance
UAS - Unmanned Aircraft System

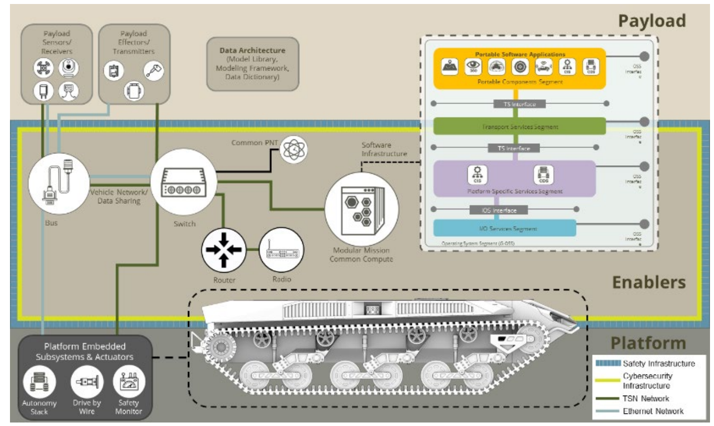
APPROVED FOR PUBLIC RELEASE

HMIF Increment 1 Baseline

Mission: Accelerate robotic and autonomous systems by integrating existing platforms and payloads and hardening them for operational employment in 27 months.



The RCCTO HMIF Pilot integrates existing systems with enhanced enablers to provide a threshold operational capability.



*Generic representation

The HMI MOSA provides the enduring open standards for Robotic and Autonomous Systems (RASs) air/ground operational integration.

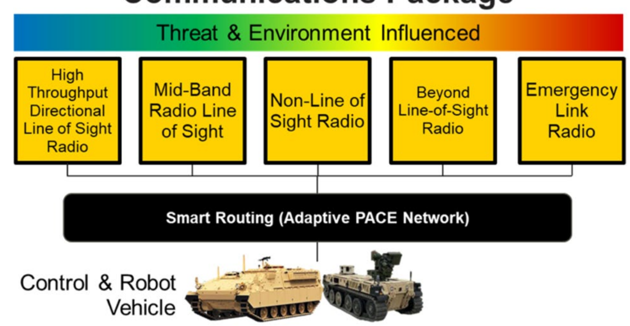
HMI Software for Autonomy and Safety

EXPERIMENTATION	OPERATIONAL INTEGRATION
Test and evaluation of novel behaviors	Application of defined behaviors
Autonomous behaviors outside established safety case	Autonomous behaviors strengthen system safety
Assumption of intervention by test/ development team	Necessity for autonomous fault recovery behaviors
Direct oversight of each behavior and sequence	Trust and safety in performance and outcome
Total human supervision of machine	Machine autonomy should lessen the burden of safety

Requires robust network, autonomy, and safety solutions that enable operational integration.

HMI Smart Routing Network

Communications Package



A resilient, low latency C2 network enables robotic control and operational mission command.

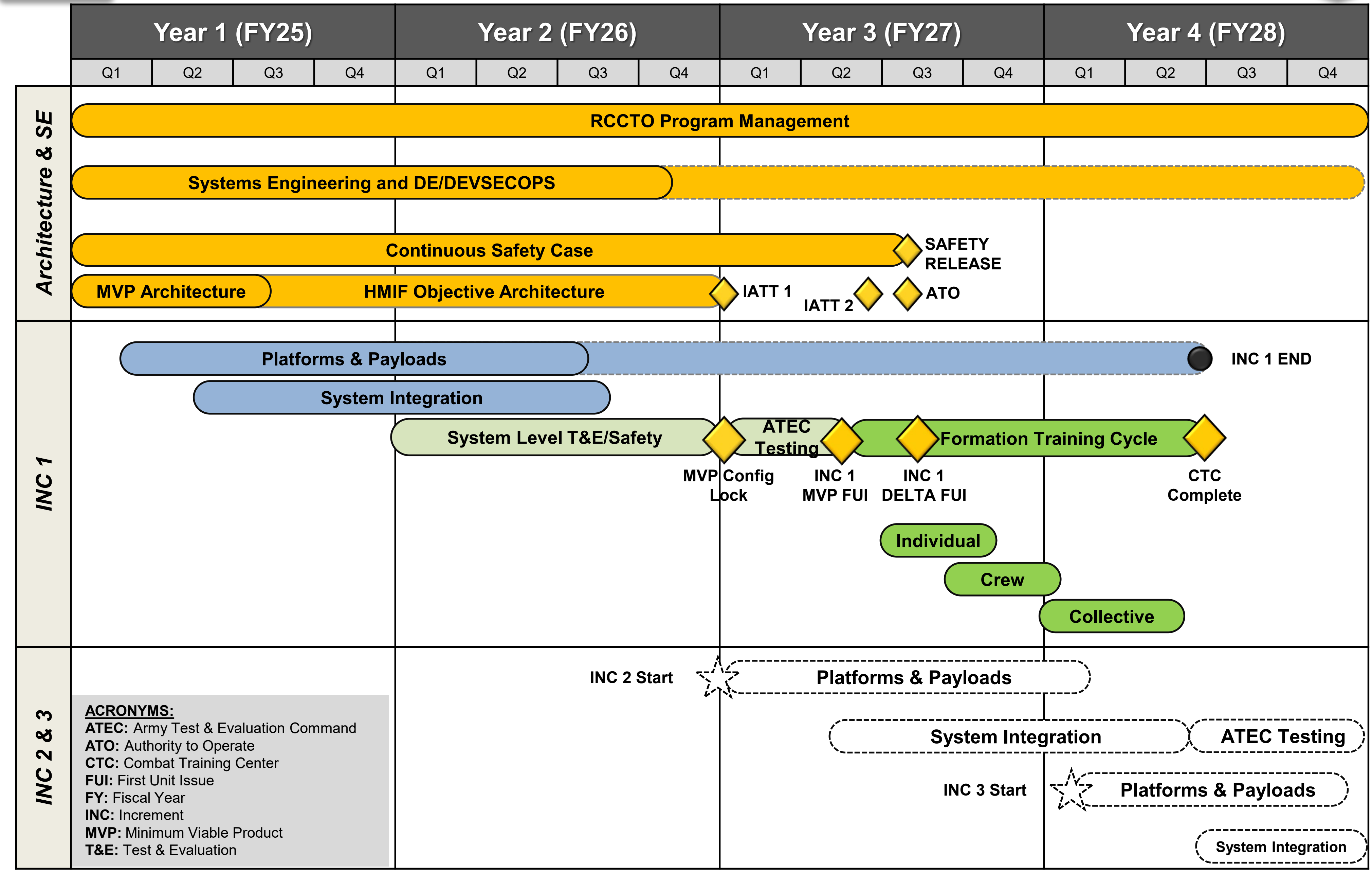
Five Outputs: Network, Common Compute, Common Controller, Safety Use Case, and Standard Hardware & Software Interface Control Documents (ICDs).

Key Advancements under HMIF

1. Advanced robotic command-and-control capabilities developed to enhance mission effectiveness.
2. A standardized integration architecture implemented to facilitate seamless interaction between robotic platforms and payloads.
3. Stringent safety protocols integrated to ensure operational safety and reliability. User interfaces optimized to enhance situational awareness and decision-making.
4. Control systems designed to be adaptable to various vehicle platforms.
5. Software applications developed to support extensible and scalable solutions.
6. Functionality expanded to support a broader range of payloads and systems.
7. Software updates incorporated to leverage the latest advancements in autonomous technologies.
8. Standard acquisition processes optimized to support accelerated development and deployment.
9. Doctrine and operational considerations refined to support emerging technologies.

Robotic capability in the hands of Soldiers.

HMIF Increment 1-3 Timeline



ACRONYMS:
ATEC: Army Test & Evaluation Command
ATO: Authority to Operate
CTC: Combat Training Center
FUI: First Unit Issue
FY: Fiscal Year
INC: Increment
MVP: Minimum Viable Product
T&E: Test & Evaluation

PM, Architecture & SE | Acquisition & Development | Government Testing | Operational w/ Unit | Future Work

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Increment 1-3 Roadmap

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INCREMENT 1 ***(FY24-27) Phase: Transformation In Contact***	***INCREMENT 2*** ***(FY27-29) Phase: Deliberate Transformation***	***INCREMENT 3*** ***(FY28-30): Phase: Deliberate Transformation***
DEFENSE RECONNAISSANCE AND SECURITY	***ATTACK (DIRECT FIRE) SUPPRESSION***	***DELIBERATE ATTACK (RECON, BREACH, PROTECT) PENETRATE***
• Area Recon • Zone Recon • Route Recon • Counter-Recon • Disrupt • Consolidate and Reorganize • Screen • Obscuration	• Conduct Support by Fire (SBF) • Conduct Local Security • Conduct Area Recon • Conduct Indirect Fire (IDF) • Conduct Attack by Fire (ABF) cooperative engagement • Suppression (Direct & Indirect Fires) • Interdict/Disrupt UAS	• Conduct Tactical Maneuver • Conduct c-UAS & Active Protection Systems • React to IDF • Conduct Suppressive Fires • Penetrate/Breach – ID Breach/Bypass • Autonomous Sustainment • Autonomous Tactical Maneuver

PAYLOADS

1. CROWS-J
2. TeUAS
3. SB600 MLL
4. C-sUAS EW
5. SOM
6. DUST
7. MAST
8. SRR/MRR

CREATION OF ADDITIONAL HMI FORMATIONS

***Formations**
Engineer Sapper PLT
Fire Support PLT

INTEGRATION OF ADDITIONAL & ENHANCED PAYLOADS

***Autonomy**
Autonomous movement to positions of advantage

***Autonomy**
Dynamic tasking and robotic teaming, voice tasking
Assured Position, Navigation & Timing

ENHANCED AUTONOMY

COMMON CONTROL

- Dismounted Control, ISV extends range
- Tele-op Control
- Dynamic Assignment
- Platform Autonomy for Complex Environments Control Network

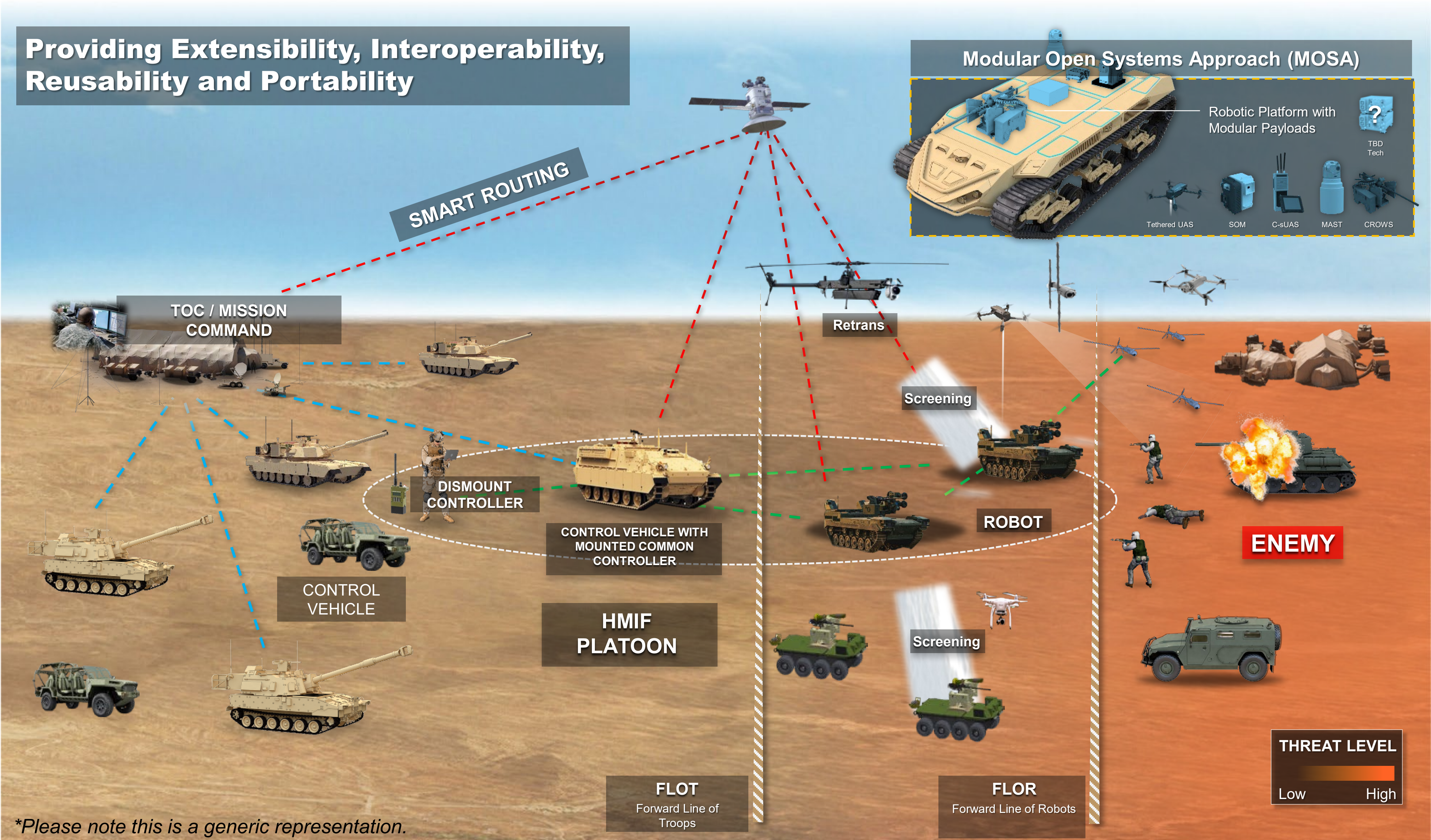
***Common Control (RAC2)**
Tactical Assault Kit (TAK) integrated C2 (platform & payload)
RAS power user concept
Edge compute

***Common Control (RAC2)**
TAK integrated C2 (expanded lethal payload control)
Improved RAS power user concept
Improved Edge compute

ENHANCED COMMON CONTROLLER CAPABILITY

***Potential capability for consideration on future increments**

HMIF Increment 1 Advancements

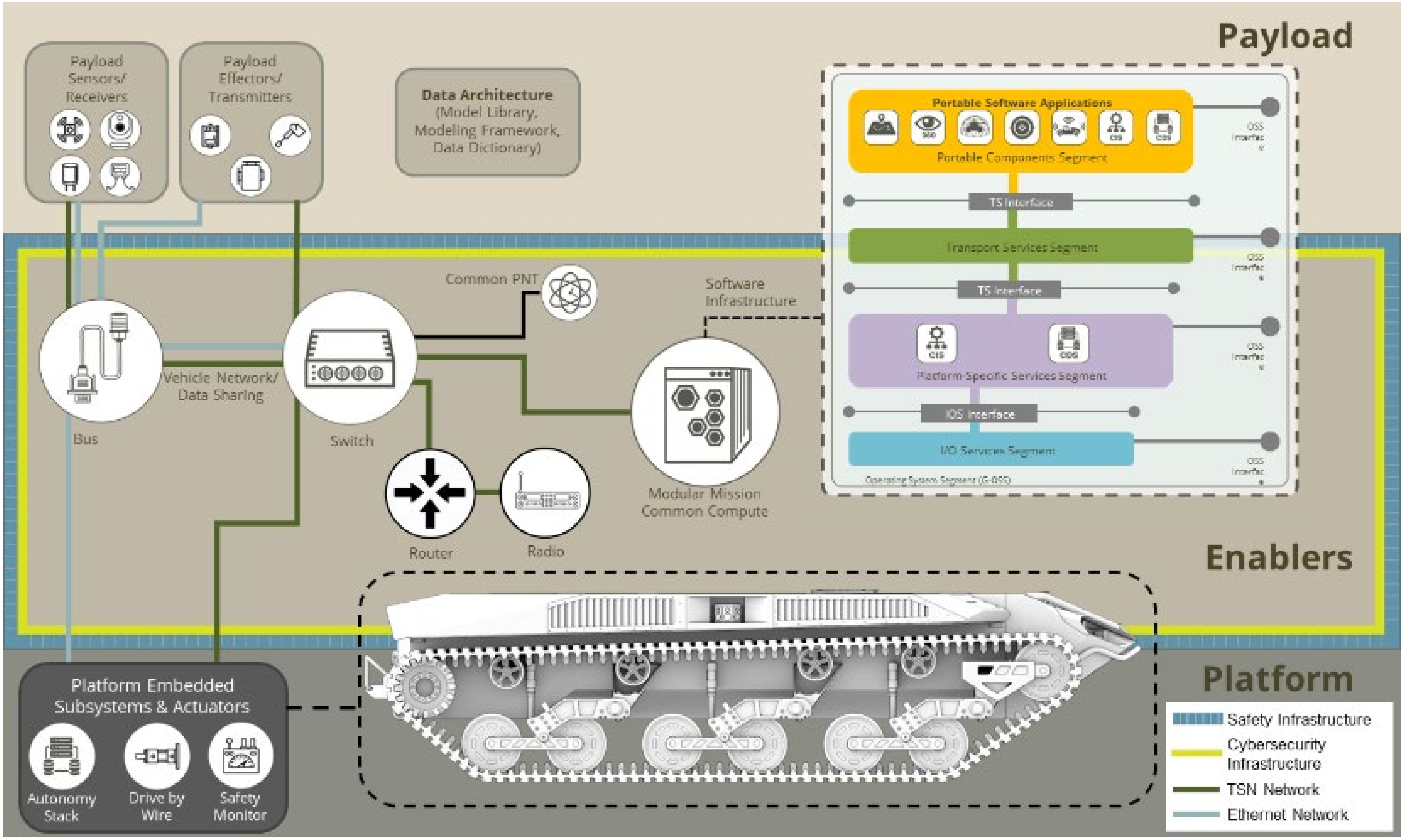


*Please note this is a generic representation.

Outputs: 1. Network; 2. Common Compute; 3. Common Controller; 4. Safety Use Case; 5. Standard Hardware & Software ICDs

HMIF SV1

- **Hierarchical approach to platform design and configuration.**
- **Platform and enablers provide infrastructure for modular, compatible payloads.**



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Concept of Architecture

Architecture is the fundamental organization of a system embodied in its elements, their relationships to each other and to the environment, and the principles guiding its design and evolution.

Architecture IS

- **Independent of device/system specific implementations:**
  - Device/ecosystem is subordinate to the architecture.
- **An enabler to achieve objective attributes and behaviors.**
- **For sharing device/subsystem level functions and information internal and external to device/subsystem.**
- **Implemented with widely used consensus based (open) standards.**
- **Identification and specification of standards for current and future devices/subsystems.**
  - Within HMIF device/system
  - Outside HMIF device/system

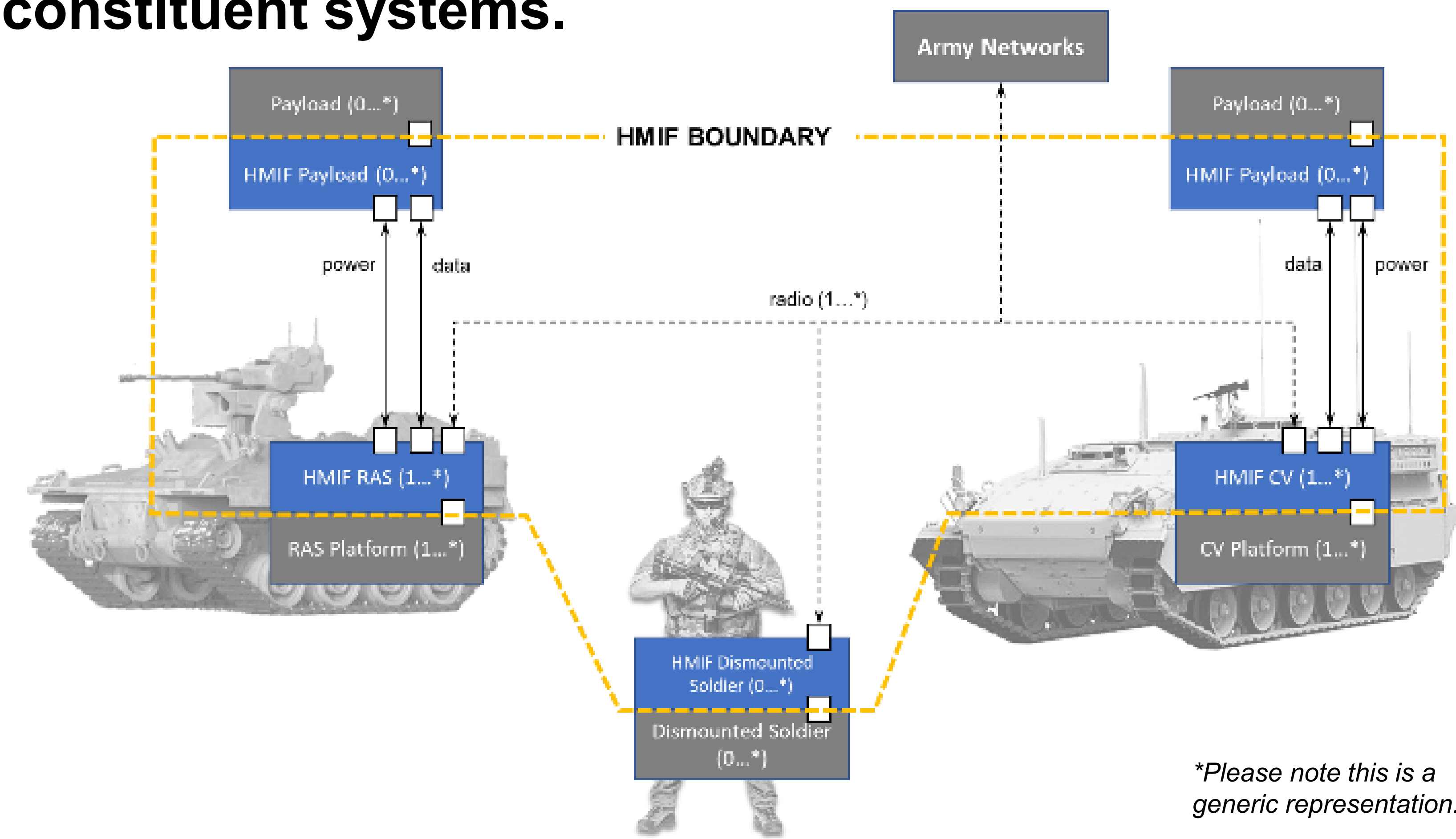
Architecture IS NOT

- **Detailed design of:**
  - Hardware
  - Electronics
  - Software
- **Definition of device/subsystem capabilities.**
- **Definition of device/subsystem internal functionality:**
  - We can treat devices/subsystems as “black boxes”.

HMIF Architecture provides guidelines that allow industry to innovate within prescribed boundaries while maintaining modularity and compatibility.

Architecture Boundary

- Architecture defines means by which systems within the SoS interface.
- Architecture does not seek to define specifics of constituent systems.



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Architecture Qualities

- **HMIF Architecture is guided by nine key qualities.**
- **Extensibility, Interoperability, Reusability, and Portability are enabled by:**
  - Modular components
  - Common components
  - Severable components
- **Controlled by common, open standards and interfaces:**
  - Consensus, industry accepted
- **Allow industry to focus on innovation with efficient, pre-defined interfaces.**

Affordability
Availability
Extensibility
Interoperability
Maintainability
Reusability &
Portability
Safety
Security
Supportability

Modular Open Systems Approach (MOSA)

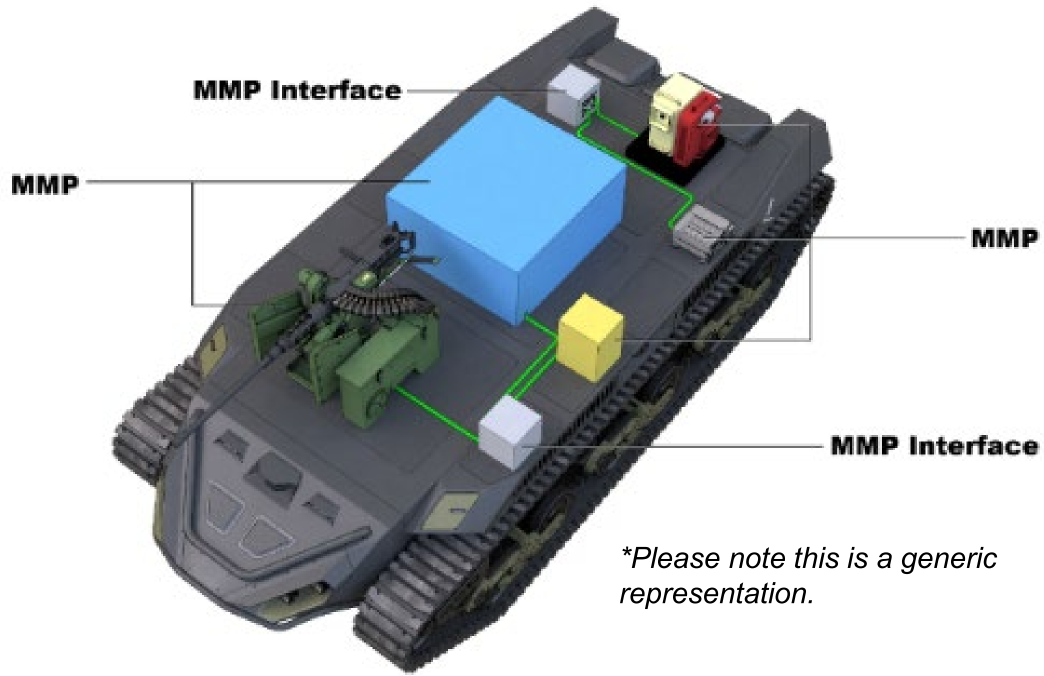
- **MOSA will allow industry to design and integrate new technologies more quickly and at reduced cost.**
- **Adherence to architecture simplifies implementation of modular components.**
- **Open standards and interfaces allow boutique and bespoke design efforts by specialized developers.**
- **Standardized data types limit systemic redundancy.**
- **MOSA limits stove-piping.**

- ***MOSA alone is not sufficient to achieve the goals of HMIF.***
- **Compatibility:**
  - Modular components utilize a minimal set of standard interfaces.
- **Commonality:**
  - Modular components reuse and share system resources.

Adherence to an established architecture will promote interoperability across a wide range of platforms, allow rapid integration and evolution of new technologies, and increase accessibility to developmental data repositories.

Modularity for HMIF

- **Modular Mission Payload (MMP):**
  - Control Vehicles
  - Robotic Platforms
- **MMP interfaces for:**
  - Power
  - Data
  - Mechanical
- **Provide open source, Interoperability Profile (IOP)-compliant interfaces for all anticipated payloads.**
- **Interchangeable payloads for rapid reconfiguration of systems based on mission:**
  - One platform, many missions



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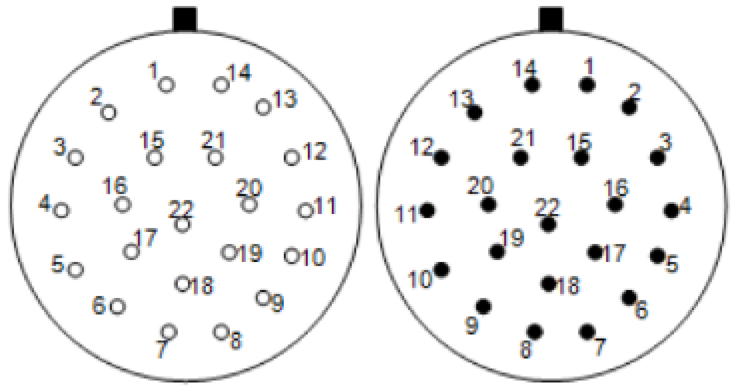
HMIF Increment 1 Payloads (anticipated)	
Tethered UAS	Remote Weapon System
Mast	Loitering Munitions
Counter Small UAS	Medium-Range Reconnaissance
Screening Obscuration Module	Short-Range Reconnaissance

Interfaces

- **Interfaces derived from open standards.**
- **IOP provides foundation of interface specifications:**
  - Mechanical
  - Electrical
  - Data
  - Software
- **Proprietary interfaces are not accepted.**
- **Interface adaptors may be used.**
- **Reduce burden on developers.**

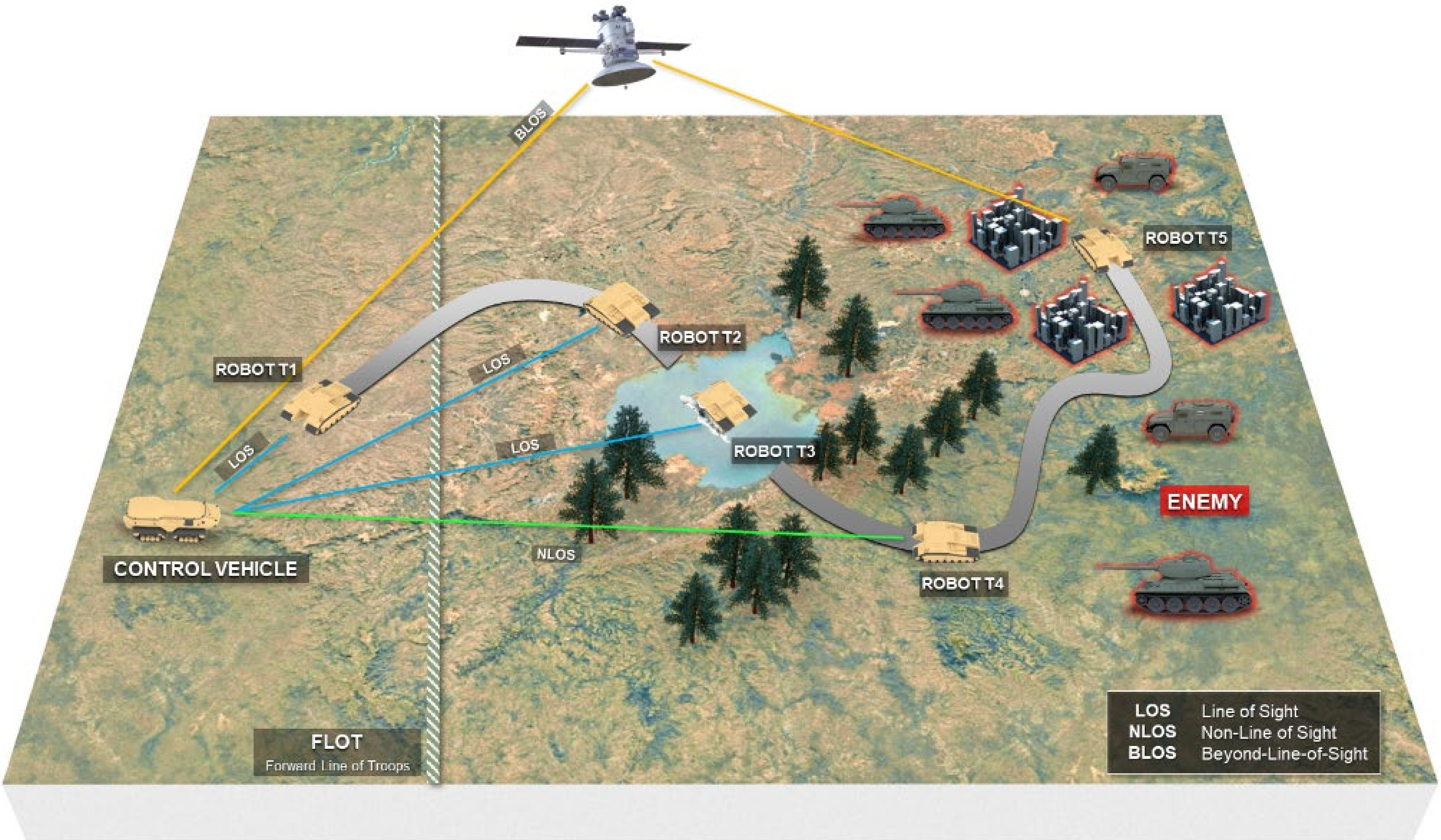


Receptacle
- Sockets
- Platform Side
- Nominal Key

Plug
- Pins
- Payload Side
- Nominal Key

Visit the National Advanced Mobility Consortium (NAMC) website for more information on pin locations and functions.

HMIF Layered Comms w/ Smart Routing Network



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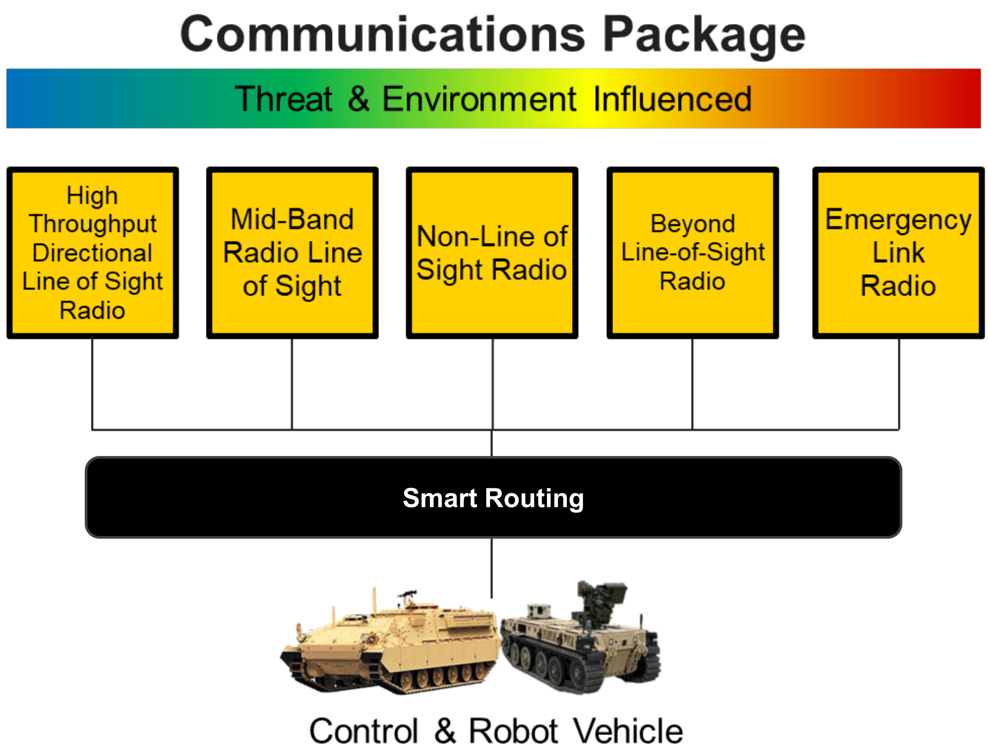
Smart Routing

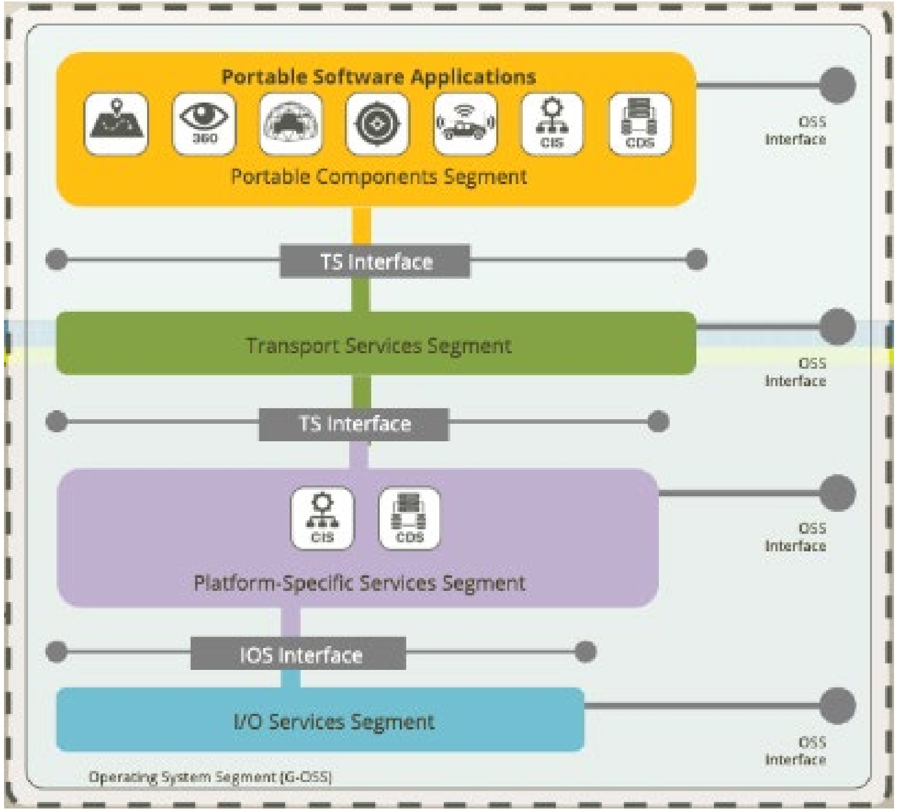


Smart Routing

- **Transport Agnostic Network:** Integrate disparate networks into a unified network, enabling traffic healing and bridging to maintain connectivity.
- **Smart Routing:** Discover, monitor, and switch between waveforms/transports/networks.
- **Traffic Conditioning:** Quality of Service (QoS)/traffic rate regulation and format translation to support disparate radio networks.
- **Increased Throughput:** Maximizes capacity utilization by load balancing traffic across multiple transports.
- **Network Optimization:** Route, gateway, and radio preset / parameter selection.
- **Commercial Networks:** Flexible to incorporate public access networks (Starlink, cellular, etc.) to provide reach (network healing) and reach back (cloud access) capabilities.
- **ITN Integration:** Seamless integration with Integrated Tactical Network to pass C2 and sensor collected information.

M2C2 Software Operating Environment

- **Modular Mission Common Compute (M2C2) and its software operating environment follow a Services Orientated Architecture (SOA). It offers a logical way of integrating complex, distributed system software by providing services via published and discoverable interfaces to applications, other services, or processes distributed across the platform network and/or wireless transport to subscribers.**



- Intelligence, Surveillance, and Reconnaissance (ISR) sensors consume services (e.g., Assured Position Navigation and Timing (APNT)) and provide services directly Cursor on Target (CoT) and indirectly through applications (Target Tracking) to subscribers on the platform or across the wireless network.
- Time-Sensitive Networking (TSN) prioritizes time-critical communication and services across available transport.
- The MOSA environment enables fast and agile capability development with reduced cost of integration, sustainment, and reuse.

HMIF Autonomy Levels

	Level One	Level Two	Level Three	Level Four	Level Five
	Operator in Control		System in Control		
	Safe			Hazardous	
	Full Operator Control: *Direct or Tele Control*	**Assisted Operator Control:** *Direct or Tele Control*	**Autonomous Operation:** *Supervised monitoring*	**Autonomous Operation:** *No Supervised monitoring*	**Autonomous Operation:** *No Supervised monitoring*
Vehicle/Navigation	No Assist Features	Example Assist Features (warnings/intervention): • Object Avoidance/ Braking • Drop/Fall Avoidance • Human Detect/Braking	Supervised human intervention/override. Continuous or Situational Assist: • Assisted maneuvers • Waypoint control • Formation Following	Continuous monitoring not required. Situational operation: control handed back to operator. Human and vehicle centered safety constraints: • Human and obstacle avoidance • Vehicle protection	Nearly unrestricted capabilities Vehicle provides safety • Friendly Centered
Active Payload	No Assist Features: Human arms and fires weapon. Fully manual payload operation.	Example Assist Features: • Target highlighting • Assisting aiming • Prevent Friendly fire • Features aimed at safety Human arms and fires weapon.	Example Features: • Autonomous target identification • Target tracking • Prevent Friendly fire Human arms and fires weapon. Supervised autonomous payload operation.	Example Features: • Target prioritization and selection. System may request to fire; human must approve. Limited Autonomous Payload operation – Safety Focused.	Human delegates approval to fire within pre-defined constraints. Human delegates ability for payload to move. Friendly fire safety constraints. Less restrictive autonomous payload operation – Friendly focused.

Increased Safety Case Requirement

Safety Engineering Architecture

Performance Claims w/ Expected Medium(-) Safety – Mode 1

Examples:

- Tethered operation
- Teleoperation
- Assisted Teleoperation
- Low Speed Supervised Autonomy (Lv2)
- Unsupervised Autonomy (Lv3/4) in Restricted Operational Domains:
  - Convoy
  - Troops away from robot (i.e., robotic operations forward of FLOT in conflict)
  - Lost Communications
- Weapon Firing
- Firing On the Move

Performance Claims w/ Expected High Safety – Mode 2

Examples:

- Some Aspects of High-Speed Supervised Autonomy (Lv2).
- Unsupervised Autonomy (Lv3/4) in Operational Domains with Intermixed Personnel:
  - Casualty Evacuation
  - Contested Logistics
  - RTB (Return to Base)
  - Lost Communications

Safety-certified interlock to allow transition to higher risk mode/allow independent assessment of system risk for each mode.

Safety Case Structure for HMIF INC 1 must enable operational learning.

Autonomy and Safety

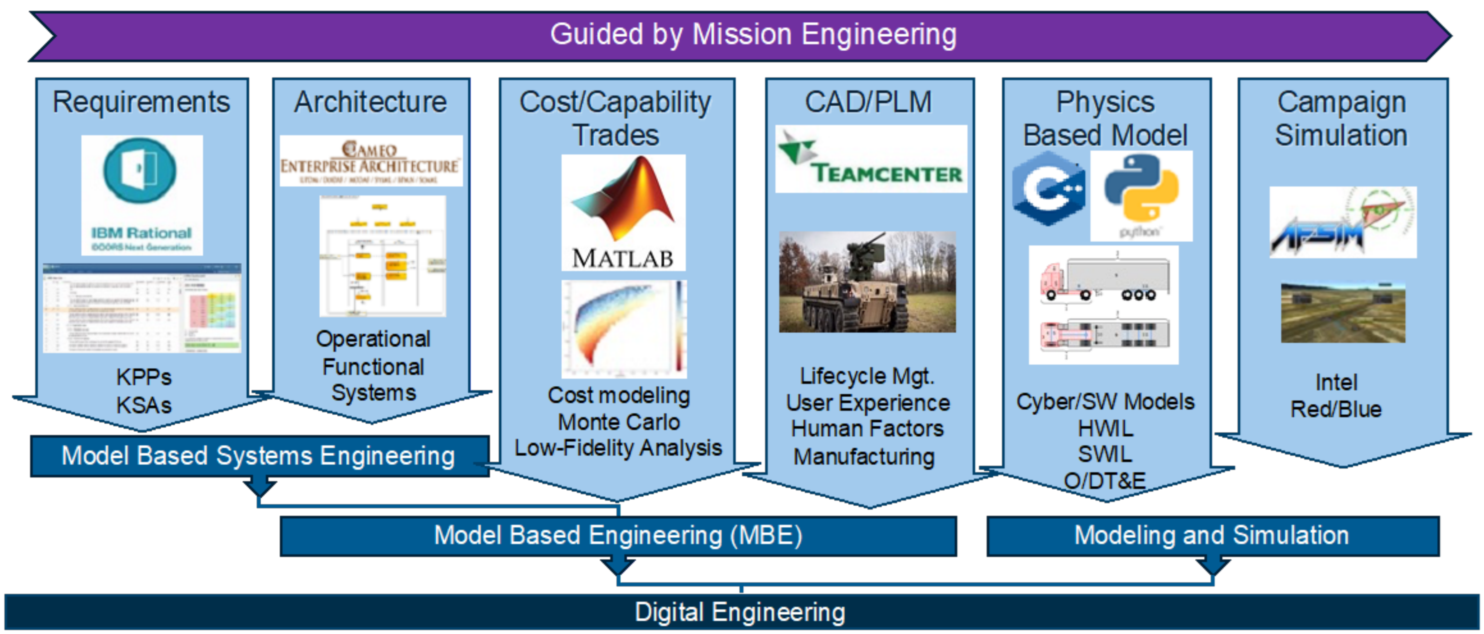
EXPERIMENTATION	*OPERATIONAL INTEGRATION*
Test and evaluation of novel behaviors.	Application of defined behaviors.
Autonomous behaviors outside established safety case.	Autonomous behaviors strengthen system safety.
Assumption of intervention by test/ development team.	Necessity for autonomous fault recovery behaviors.
Direct oversight of each behavior and sequence.	Trust and safety in performance and outcome.
Total human supervision of machine.	Machine autonomy should lessen the burden of safety.

Only a robust and mutually supportive Network, Autonomy, and Safety solution will enable Operational Integration.

Digital Engineering

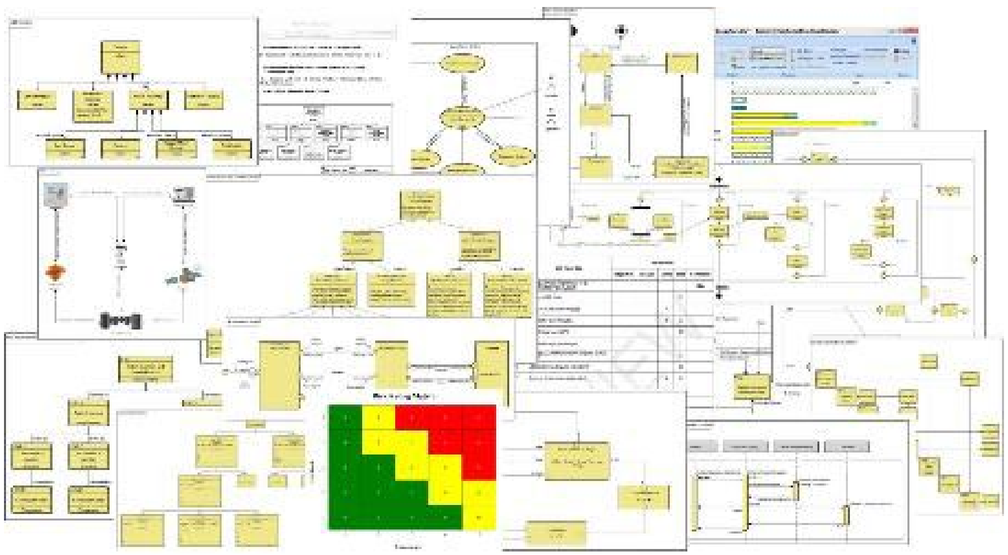
- **Strong models critical to understanding overall SoS.**
- **SysML used to model all SoS components:**
  - SysML model as authoritative source of truth.
  - All other documentation subordinate to model.
- **Vendor deliverables will include digital artifacts:**
  - SysML models
  - Computer-aided design/engineering

- **Collaboration Environment:**
  - Provides Community of Interest (COI) access to SoS models.
  - Interface Description Document (ICD):
    - ICDs will be derived from Interface Design Description (IDD).
  - Open Systems Management Plan.
  - Architecture Description Document:
    - Aspects of model to be available to community of interest.



Verification, Validation, and Testing

- SysML model is tool for early verification and validation (V&V) of interface and functionality.
- Vendors will develop product models.
- Vendors will have access to Government-maintained SoS model to conduct V&V.

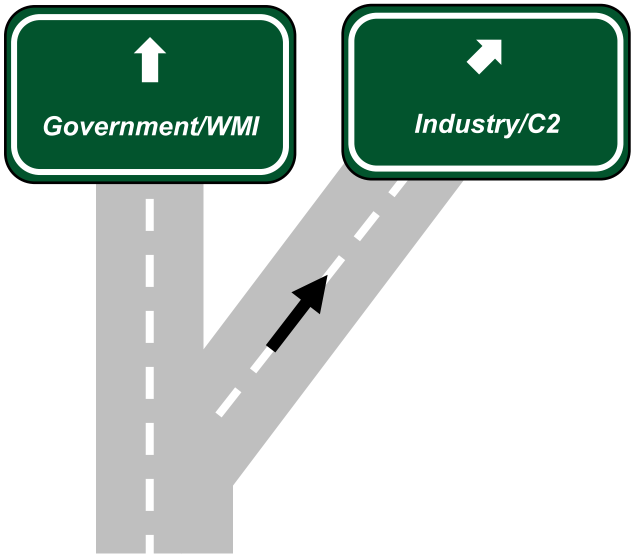


*Please note this is a generic representation and not the HMIF V&V Test architecture.

- System Integration Lab (SIL) will be developed and maintained by Govt.
- Government will evaluate performance of systems within the SoS.
- ATEC to perform functional and safety testing.
- <u>Operational Prototype</u> will be delivered to designated Army units for extended evaluation.

HMIF Command and Control Software Pivot

<u>Rationale</u>: The Government's Warfighter-Machine Interface (WMI) is not aligned to future C2 efforts. RCCTO will work with industry to develop C2 software for robotic and unmanned vehicles by bringing in Industry to leverage their capabilities and create the necessary C2 software components.



Benefits:

- Architecture that scales with Army requirements.
- Reduced need for software development.
- Facilitates path for rapid acquisition.
- Greater alignment w/Enterprise C2 objectives.

Path Forward:

- Collect Industry feedback and allow Industry opportunity to “self-form” into teams to address the C2 software problem set.
- Incorporate industry feedback to further inform our C2 software requirement.

Government framework, Industry development

How Can Industry Help?

We need the best of Industry to build to Government architecture.